Firefly Shipping Model

Due Date: 11:59 p.m. EST, Sunday of Unit 6
Points: 100

Overview:

Mal Foil the owner of Firefly Shipping has asked you to create a model to project the number of units they must be prepared to ship in the first month of the new year. Mal has supplied you with a template of what he would like to see.

Instructions:

- Download: Unit 06 Assignment – Firefly Shipping Model.xlsx
- Mal also has a specific set of requirements:
  - use a 3-month weighted average to forecast January.
  - see how the 3-month weighted average method would perform against historical data.
  - see the 3-month weighted average charted out within the model.
  - Mal is not sure how what weights he would like to use. Your model should allow him to adjust the weights, listed in G4-I4, and see these changes change the models output.

- The 3-month weighted average formula is listed below:

$$F_t = \frac{(W_{t-1} * U_{t-1}) + (W_{t-2} * U_{t-2}) + (W_{t-3} * U_{t-3})}{(W_{t-1} + W_{t-2+W_{t-3}})}$$

  - F_t = The forecasted values (column D)
  - W_{t-X} = the weight for the prior months. For example, if F_t = April then, W_{t-1} would equal to March, W_{t-2} would equal February and W_{t-3} would equal January.
  - U_{t-X} = the units for the prior months, similar to how the weights work.

Requirements:

- Microsoft Excel file

Be sure to read the criteria by which your work will be evaluated before you start and again after before you submit.

Evaluation Rubric for *Firefly Shipping Model* Assignment

CRITERIA	Deficient	Needs Improvement	Proficient	Exemplary
	(0-4 Points)	**(16-18 Points)**	**(20-24 Points)**	**(25 Points)**
3-Month WMA	Missing, or incomplete.	Complete but showing inaccurate or static formula.	Complete and showing all or mostly correct.	All formulas are correct, formatting has been applied to outputs.
Weights Adjusted	Missing, or incomplete.	Complete but showing inaccurate or static formula.	Complete and showing all or mostly correct.	All formulas are correct, formatting has been applied to outputs.
	(0-29 Points)	**(30-39 Points)**	**(40-49 Points)**	**(50 Points)**
Chart	Graph missing or showing the wrong data.	Graph has been included, but not well labeled and may be showing the incorrect data.	Graph presents the data in a readable format, adequate labeling.	Graph is clear, data is shown and is correct. All axes have been labeled with correct number formatting.